Documentation of the Hidden Sector Dark Matter Toolkit

Code Structure, Boltzmann Equations, and Assumptions

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Contents

1 Introduction **1**

2 Cosmological Background **1**
2.1 Standard Model Thermal Bath and Hubble Rate 1
2.2 Equilibrium Distributions and Energy Moments 2

3 Hidden Sector Models **3**
3.1 `VectorPortal`: Kinetic Mixing with Hypercharge 3
3.1.1 Hidden-sector processes 4
3.1.2 Connection to the Standard Model 5
3.2 `BLPortal`: $B - L$ Kinetic Mixing 6
3.3 `BaryonPortal`: Baryon Number Kinetic Mixing 7
3.4 `LiLjPortal`: $L_i - L_j$ Kinetic Mixing 7
3.5 `HiggsPortal`: Scalar Mixing 8
3.5.1 Hidden-sector processes 8
3.5.2 Connection to the Standard Model 9

4 Evolution of the Hidden Sector **11**
4.1 Boltzmann Equations 11
4.1.1 Reformulation in terms of chemical potentials and ξ 12
4.1.2 Solver Implementation 13
4.2 Late-time background evolution and mediator decay 14

5 Mapping the Portal Coupling: Predictions and Constraints **17**
5.1 Dark Matter Relic Abundance 17
5.2 Constraints from Big Bang Nucleosynthesis 18
5.3 Constraints from Direct Detection 19
5.4 Thermalization Floor 20

A Additional Approximations and Validity **20**

B Details on the Numerical Implementation and Alternative Solvers **22**
B.1 Alternative solvers 22
B.2 Numerical settings 22

C Derivation of the Boltzmann system in Chemical-potential Variables **23**

D Physical Constants **26**

E Code Architecture **27**

References **29**

1 Introduction

This document provides a detailed technical description of the `hidden_sector_DM` Python toolkit[1], which computes the relic abundance of dark matter (DM) in *hidden sector* scenarios. In these models, the DM particle X annihilates primarily into unstable mediator particles Y, which subsequently decay to Standard Model (SM) final states through a small portal coupling ε. The defining feature is the parametric hierarchy

$$g_D \gg \varepsilon\,, \tag{1.1}$$

where g_D denotes the hidden-sector coupling strength, so that the DM annihilation rate $X\bar{X} \to YY$ is set by the hidden-sector interactions while the mediator decay rate $Y \to \text{SM}$ is controlled by the much smaller portal coupling.

The code solves the coupled Boltzmann equations for the DM yield Y_X, the mediator yield Y_Y, and the dark-to-SM temperature ratio $\xi = T_h/T_{\rm SM}$, incorporating:

- $2 \to 2$ annihilation: $X\bar{X} \leftrightarrow YY$
- $3 \to 2$ cannibal processes: $YYY \to YY$, $YYX \to YX$, $YXX \to XX$
- Mediator decay: $Y \to \text{SM}$ with portal coupling ε
- Hidden sector heating and entropy injection

Five portal models are implemented: kinetic mixing with hypercharge (`VectorPortal`), $B-L$ gauge boson mixing (`BLPortal`), $L_i - L_j$ gauge boson mixing (`LiLjPortal`), baryon-number gauge boson mixing (`BaryonPortal`), and scalar mixing with the SM Higgs boson (`HiggsPortal`). Observational constraints from the DM relic abundance, Big Bang Nucleosynthesis (BBN) hadronic injection bounds, and direct detection experiments (particularly the combination of LZ and XENONnT bounds) are computed for each model.

2 Cosmological Background

This section establishes the cosmological quantities that enter the Boltzmann equations of Sec. 4.1. We first define the Standard Model (SM) thermal bath and the Hubble rate, including the modification from the hidden-sector energy density, and then collect the equilibrium distributions and energy moments under the Maxwell–Boltzmann approximation that are used throughout the solver.

2.1 Standard Model Thermal Bath and Hubble Rate

The SM thermal background is described by the relativistic degrees of freedom $g_*(T)$ and $g_{*S}(T)$, loaded from tabulated data files (covering $T \sim 0.1\,\text{MeV}$ to $\sim 100\,\text{GeV}$) and interpolated linearly

[1] https://github.com/GabrieleMonte/hidden_sector_DM.

in T. The key thermodynamic quantities are:

$$\rho_{\rm rad}(T) = \frac{\pi^2}{30}\, g_*(T)\, T^4 \qquad \text{(radiation energy density)} \tag{2.1}$$

$$s(T) = \frac{2\pi^2}{45}\, g_{*S}(T)\, T^3 \qquad \text{(entropy density)} \tag{2.2}$$

$$H(T) = \sqrt{\frac{\pi^2}{90}\, g_*(T)}\; \frac{T^2}{M_{\rm Pl}} \qquad \text{(Hubble rate, SM only)} \tag{2.3}$$

where $M_{\rm Pl} = 2.435 \times 10^{18}\,\text{GeV}$ is the reduced Planck mass. The factor

$$\tilde{g}(T) = 1 + \frac{1}{3}\frac{d\ln g_{*S}}{d\ln T} \tag{2.4}$$

accounts for the temperature dependence of g_{*S} in the entropy conservation equation ($\tilde{g} = 1$ when g_{*S} is constant).

When the hidden sector contributes non-negligibly to the total energy density, the Hubble rate is modified:

$$H_{\rm tot} = \sqrt{H^2 + \frac{\rho_h}{3M_{\rm Pl}^2}}\,, \qquad \rho_h = \rho_X + \rho_Y. \tag{2.5}$$

2.2 Equilibrium Distributions and Energy Moments

All equilibrium quantities use the Maxwell-Boltzmann approximation (see Sec. A for a discussion of its validity). For a species with mass m, internal degrees of freedom g, at a temperature T, the number density and its derivative are given by:

$$n_{\rm MB}^{\rm eq}(z) = \frac{g\, m^3}{2\pi^2}\,\frac{K_2(z)}{z}\,, \qquad \frac{dn_{\rm MB}^{\rm eq}}{dT} = \frac{g\, m^2}{2\pi^2}\Big[z\, K_1(z) + 3\, K_2(z)\Big] \tag{2.6}$$

where $z \equiv m/T$ and $K_\nu(z)$ is the modified Bessel functions of the second kind. A useful combination is the logarithmic derivative

$$\lambda(z) \equiv \frac{d\ln n_{\rm MB}^{\rm eq}}{d\ln T} = z\,\frac{K_1(z)}{K_2(z)} + 3\,, \tag{2.7}$$

which appears in the Boltzmann equations whenever equilibrium tracking is needed.

Higher moments of the distribution give the energy density and pressure,

$$\rho = B_1\, n_{\rm MB}^{\rm eq}\,, \qquad \rho + P = B_2\, n_{\rm MB}^{\rm eq}\,, \tag{2.8}$$

where

$$B_1(z,m) = m\left(\frac{K_1(z)}{K_2(z)} + \frac{3}{z}\right), \qquad B_2(z,m) = m\left(\frac{K_1(z)}{K_2(z)} + \frac{4}{z}\right) \tag{2.9}$$

which can be thought of as the average energy per particle, $B_1 = \langle E\rangle/n$, and the enthalpy per particle, $B_2 = (\rho + P)/n$, respectively. Note that the difference $B_2 - B_1 = m/z = T$ recovers the ideal-gas relation $P = nT$. Their temperature derivatives are

$$\frac{dB_1}{dT} = 3 - z^2\left[R^2 - 1 + \frac{3R}{z}\right], \tag{2.10}$$

$$\frac{dB_2}{dT} = 4 - z^2\left[R^2 - 1 + \frac{3R}{z}\right], \tag{2.11}$$

where $R \equiv K_1(z)/K_2(z)$.

Another useful quantity that is implemented in the Boltzmann equations is the logarithmic equilibrium yield, $\ln Y_i^{\rm eq} = \ln n_{\rm MB}^{\rm eq} - \ln s$, with $\ln 2$ added if antiparticles are included.

Code implementation: asymptotic expansions. In our code, the density is set to zero for $z > 500$ to avoid underflow and is doubled when antiparticles are present. For $z \geq 300$, the equilibrium number density is computed via the asymptotic form

$$\ln n_{\rm MB}^{\rm eq} = \ln\Big(\frac{g\, m^3}{2\pi^2}\Big) + \frac{1}{2}\ln\frac{\pi}{2z} - z + \ln\Big(1 + \frac{15}{8z} + \frac{105}{128z^2}\Big) - \ln z\,. \tag{2.12}$$

The Bessel ratio $R = K_1(z)/K_2(z)$, which enters B_1, B_2, λ, and their derivatives, is computed directly from `scipy.special.kv` for $z < 50$ while for $z \geq 50$ the following expansion is used instead:

$$\frac{K_1(z)}{K_2(z)} \approx 1 - \frac{3}{2z} + \frac{15}{8z^2} - \frac{105}{16z^3}. \tag{2.13}$$

In the same regime, the bracket in the B-factor derivatives simplifies to $3/(2z^2) - 15/(4z^3)$.

Full quantum statistics (`species_thermo`): A separate method computes (ρ, P, n) with full Fermi-Dirac or Bose-Einstein integrals using a three-regime approach (ultra-relativistic analytic for $z \leq 10^{-3}$, numerical integral for $10^{-3} < z < 1$, MB for $z \geq 1$). This is available for auxiliary calculations but is *not* used in the Boltzmann solver itself.

3 Hidden Sector Models

This section presents the five portal models implemented in the code. Each model specifies the hidden-sector particle content (masses m_X and m_Y, internal degrees of freedom g_X g_Y, and whether antiparticles are included), the $2 \to 2$ annihilation and $3 \to 2$ cannibal cross sections that govern the freeze-out dynamics, the mediator decay width $\Gamma(Y \to \text{SM})$, and the spin-independent DM–nucleon scattering cross section $\sigma_{\rm SI}$. All models inherit from the abstract base class `DarkSectorModel`, which defines this common interface, and differ only in how the portal coupling ε connects the hidden sector to the Standard Model. The initial dark-to-SM temperature ratio $\xi_{\rm ini}$ is a free parameter common to all models.

3.1 `VectorPortal`: Kinetic Mixing with Hypercharge

The DM particle X is a Dirac fermion ($g_X = 2$, includes antiparticles) charged under a dark $U(1)_D$ gauge symmetry. The mediator $Y \equiv Z'$ is the associated dark photon ($g_Y = 3$, massive spin-1, no antiparticle doubling). The dark photon mixes kinetically with the SM hypercharge gauge boson via the mixing parameter ε. We first describe the hidden-sector processes — $2 \to 2$ annihilations and $3 \to 2$ cannibal reactions — and then the connection to the visible sector through kinetic mixing, which controls both the mediator decay width and the DM–nucleon scattering cross section.

3.1.1 Hidden-sector processes

The thermally averaged $2 \to 2$ annihilation cross section for $X\bar{X} \to Z'Z'$ is

$$\langle \sigma v \rangle_{X\bar{X}\to Z'Z'} = 2\left(a + \frac{6b}{x_X}\right), \tag{3.1}$$

where $x_X = m_X/T_h$, $r_v \equiv m_Y/m_X$, and

$$a = \frac{2\pi\alpha_X^2}{m_X^2} \frac{(1-r_v^2)^{3/2}}{(2-r_v^2)^2}, \tag{3.2}$$

$$b = \frac{\pi\alpha_X^2}{12\, m_X^2} \frac{\sqrt{1-r_v^2}\,(24 + 28r_v^2 - 36r_v^4 + 17r_v^6)}{(2-r_v^2)^4} \tag{3.3}$$

are respectively the s-wave and p-wave coefficients. For the background evolution solver, where the p-wave contribution is subdominant at late times ($x \gg 1$), only the s-wave piece $\langle \sigma v \rangle_{\text{s-wave}} = 2a$ is retained.

Three $3 \to 2$ cannibal processes contribute to number-changing interactions within the hidden sector. The dominant channel for the freeze-out of cannibal processes is $YYX \to YX$, whose thermally averaged cross section

$$\langle \sigma v^2 \rangle_{YYX\to YX} = \Delta_3 \frac{\alpha_X^3}{m_X^5} \tag{3.4}$$

is computed at tree level with `FeynCalc` [**? ? ? ?**] in the non-relativistic limit, which is the regime of interest since these rates only begin to affect the dynamics once Y starts to depart from chemical equilibrium. The resulting coefficient is

$$\Delta_3 = \frac{4\pi^2\, r^5 \sqrt{\frac{4}{3}r(r+2)+1}\, \mathcal{N}_3(r)}{(r+2)^3\,(2r+1)^4\left[2r^2(r+2)+r-1\right]^2} \tag{3.5}$$

where $r = m_X/m_Y$ and

$$\mathcal{N}_3(r) = 2r\Big(r\big(r(r(2r(r(8r(10r+71)+1683)+2798)+6107)+4722)+2335\big)+578\Big)+195\,. \tag{3.6}$$

The remaining two channels are [1]:

$$\langle \sigma v^2 \rangle_{YYY\to YY} = \Delta_1 \frac{\alpha_X^5\, T_h^7}{m_X^{12}}, \tag{3.7}$$

$$\langle \sigma v^2 \rangle_{YXX\to XX} = \Delta_2 \frac{\alpha_X^3}{m_X^5}, \tag{3.8}$$

where Δ_2 is also computed at tree level with `FeynCalc` in the same non-relativistic limit:

$$\Delta_2 = \frac{\pi^2\, r^3\,(1+4r)^{3/2}\, \mathcal{N}_2(r)}{6\,(1+2r)^3\,(2r^2+r-1)^2} \tag{3.9}$$

with

$$\mathcal{N}_2(r) = 256r^6 - 384r^5 + 536r^4 - 96r^3 + 38r^2 + 72r + 31\,. \tag{3.10}$$

The $YYY \to YY$ rate is loop-suppressed relative to the other two channels (α_X^5 vs. α_X^3) and carries an explicit temperature dependence, making it negligible for the evolution of the hidden sector. We retain it for completeness with a phenomenological coefficient $\Delta_1 = 0.5$.

3.1.2 Connection to the Standard Model

After restoring canonical kinetic terms, the Z' acquires couplings to SM fermions proportional to ε. For a fermion f with electric charge Q_f and weak hypercharges Y_L, Y_R, the left- and right-handed couplings are [1]

$$g_L = \varepsilon \, \frac{m_{Z'}^2 \, g_Y^{\rm SM} \, Y_L - M_Z^2 \, g_W \, s_W \, c_W \, Q_f}{M_Z^2 - m_{Z'}^2} \,, \tag{3.11}$$

$$g_R = \varepsilon \, \frac{m_{Z'}^2 \, g_Y^{\rm SM} \, Y_R - M_Z^2 \, g_W \, s_W \, c_W \, Q_f}{M_Z^2 - m_{Z'}^2} \,, \tag{3.12}$$

following [**?** **?**], where g_W and $g_Y^{\rm SM}$ are the $SU(2)_L$ and $U(1)_Y$ gauge couplings, $s_W \equiv \sin\theta_W$ and $c_W \equiv \cos\theta_W$ with θ_W the Weinberg angle, and M_Z is the Z-boson mass. The vector and axial couplings to each fermion species are then

$$g_{fV} = \frac{g_R + g_L}{2} \,, \qquad g_{fA} = \frac{g_R - g_L}{2} \,. \tag{3.13}$$

Mediator decay width. These couplings determine the Z' decay width. The partial width into a fermion pair is

$$\Gamma(Z' \to f\bar{f}) = \frac{N_c^f \, m_{Z'}}{12\pi} \sqrt{1 - r_f^2} \left[g_{fV}^2 \left(1 + \tfrac{1}{2} r_f^2\right) + g_{fA}^2 \left(1 - r_f^2\right) \right] \tag{3.14}$$

where $r_f = 2m_f / m_{Z'}$ and N_c^f is the colour factor (3 for quarks, 1 for leptons). The total width is the sum over all kinematically accessible channels ($m_{Z'} > 2m_f$):

$$\Gamma_{Z'}^{\rm tot} = \sum_{f \,:\, m_{Z'} > 2m_f} \Gamma(Z' \to f\bar{f}) \,. \tag{3.15}$$

Direct-detection cross section. The same kinetic mixing generates a tree-level DM–nucleon elastic scattering amplitude, mediated by the coherent exchange of the two mass eigenstates Z and Z' after electroweak symmetry breaking [2**?**]. In the parameter space of interest here, namely $m_{Z'}/m_X \gtrsim 0.1$, recoil-energy effects on the scattering are negligible [**?**], and the cross section reads simply in terms of the Z–Z' mixing angle η defined by [2]

$$\tan(2\eta) = \frac{2\,\varepsilon\, s_W}{1 - m_{Z'}^2 / M_Z^2} \,, \tag{3.16}$$

and the DM couplings to the mass eigenstates are $V_\chi^Z = g_D \sin\eta$ and $V_\chi^{Z'} = g_D \cos\eta$, with $g_D = \sqrt{4\pi\alpha_X}$. The quark couplings receive contributions from both the electromagnetic and weak neutral currents:

$$V_q^Z = \sqrt{4\pi\alpha_{\rm EM}}\,\varepsilon\, \sin\eta\, Q_q + \frac{\sqrt{4\pi\alpha_{\rm EM}}}{s_W c_W} \left(\cos\eta + \varepsilon\, s_W\, \sin\eta\right) \left(T_3^q - s_W^2\, Q_q\right) , \tag{3.17}$$

$$V_q^{Z'} = \sqrt{4\pi\alpha_{\rm EM}}\,\varepsilon\, \cos\eta\, Q_q + \frac{\sqrt{4\pi\alpha_{\rm EM}}}{s_W c_W} \left(\sin\eta - \varepsilon\, s_W\, \cos\eta\right) \left(T_3^q - s_W^2\, Q_q\right) , \tag{3.18}$$

where $\alpha_{\rm EM} \simeq 1/137$ is the electromagnetic fine-structure constant, Q_q the electric charge of quark q, and T_3^q its weak isospin. The spin-independent cross section per nucleon is then

$$\sigma_{\rm SI} = \frac{\mu^2}{\pi} \left[\sum_{i \in \{Z, Z'\}} \frac{V_\chi^i \left[\mathcal{Z}(2V_u^i + V_d^i) + (\mathcal{A} - \mathcal{Z})(V_u^i + 2V_d^i) \right]}{\mathcal{A}\, m_i^2} \right]^2 \times \hbar^2 c^2 \tag{3.19}$$

where $\mu = m_X\, m_N/(m_X + m_N)$ is the reduced mass with $m_N = (m_p + m_n)/2$ the average nucleon mass, and $\mathcal{Z}$ and $\mathcal{A}$ are the atomic and mass numbers of the target nucleus; the explicit form of Eq. (3.19) follows [2**?** **?**].

Two notable features appear in the cross section as a function of $m_{Z'}$. First, the mixing angle of Eq. (3.16) diverges at $m_{Z'} = M_Z$, causing the vector couplings $V_q^{Z,Z'}$ to blow up on resonance; this pole would be regularised by the finite width $\Gamma_{Z'}$, which we neglect here, but the resulting enhancement of $\sigma_{\rm SI}$ remains sizeable in the vicinity of the Z pole. Second, the coherent sum of Z and Z' exchange admits an exact cancellation below the pole, where the two amplitudes interfere destructively. For a xenon target, this blind spot occurs at

$$m_{Z'}^{\not 0} \simeq 64.3 \text{ GeV} \;\; (\approx 0.705\, M_Z) \qquad \text{(Xe-131 target)}, \tag{3.20}$$

at which $\sigma_{\rm SI}$ vanishes identically, to leading order in ε. The location of this zero depends on the target nucleus through the $(\mathcal{Z}, \mathcal{A})$-weighted quark couplings in Eq. (3.19), but not on any hidden-sector parameter.

The default parameter values used in the `VectorPortal` class are summarised in Table 1.

Parameter	Symbol	Default
DM mass	m_X	100 GeV
Mediator mass	m_Y	30 GeV
Dark coupling	α_X	10^{-2}
DM DOF	g_X	2 (Dirac fermion)
Mediator DOF	g_Y	3 (massive vector)
Cannibal rescaling	Δ_1	0.5
Cannibal rescaling	Δ_2, Δ_3	exact tree-level
Initial temperature ratio	$\xi_{\rm ini}$	1.0

Table 1: Default parameters for the `VectorPortal` model.

3.2 `BLPortal`: $B - L$ Kinetic Mixing

The hidden-sector Z' mixes kinetically with the $U(1)_{B-L}$ gauge boson Z_{B-L}. The DM particle X is a Dirac fermion coupled to the Z' as in the `VectorPortal`, so the hidden-sector dynamics — annihilation $X\bar{X} \to Z'Z'$ and the three $3 \to 2$ cannibal channels — are identical to those of Sec. 3.1 and the $2 \to 2$ and $3 \to 2$ cross sections of Eqs. (3.2)–(3.4) carry over unchanged. In contrast to the hypercharge portal, however, the kinetic mixing induces *purely vectorial* couplings of the Z' to SM fermions [3]:

$$g_{fV} = \varepsilon\, g_{B-L}\, (B-L)_f \left| \frac{m_{Z_{B-L}}^2 + m_{Z'}^2}{m_{Z_{B-L}}^2 - m_{Z'}^2} \right|, \qquad g_{fA} = 0\,, \tag{3.21}$$

where g_{B-L} is the $U(1)_{B-L}$ gauge coupling, $m_{Z_{B-L}}$ the corresponding gauge boson mass, and $(B-L)_f = +1/3$ for quarks and -1 for leptons. Throughout this work we adopt $m_{Z_{B-L}} = 10\,\text{TeV}$, sufficiently above the Z' mass range of interest that the kinematic factor in Eq. (3.21) is effectively

unity and the coupling reduces to $g_{fV} \approx \varepsilon\, g_{B-L}\, (B-L)_f$. The Z' decay width retains the functional form of Eq. (3.14), evaluated with $g_{fA} = 0$ and summed over all kinematically accessible SM fermions, including leptons.

For direct detection, the universal $(B-L)_q = 1/3$ quark charge makes the resulting spin-independent DM–nucleon cross section isospin-universal, with protons and neutrons receiving equal contributions:

$$\sigma_{\rm SI} \approx \frac{4\,\mu^2\,\alpha_X\,\varepsilon^2\,g_{B-L}^2}{m_{Z'}^4}\,, \tag{3.22}$$

where μ is the DM–nucleon reduced mass. Unlike the hypercharge and Higgs portals, only a single mediator contributes here, so no blind spot arises.

3.3 `BaryonPortal`: Baryon Number Kinetic Mixing

The hidden-sector Z' mixes kinetically with the $U(1)_B$ gauge boson Z_B [**? ? ?**]. This scenario is not to be confused with baryon-portal models involving four-fermion operators of the form *udd*. As in the $B-L$ case, X is a Dirac fermion coupled to the Z' exactly as in the `VectorPortal`, so the hidden-sector dynamics ($X\bar{X} \to Z'Z'$ and the three $3 \to 2$ cannibal channels) carry over unchanged from Sec. 3.1. Kinetic mixing induces purely vectorial couplings of the Z' to SM fermions, but now limited to quarks alone [3]:

$$g_{fV} = \varepsilon\, g_B\, B_f\, \left|\frac{m_{Z_B}^2 + m_{Z'}^2}{m_{Z_B}^2 - m_{Z'}^2}\right|, \qquad g_{fA} = 0\,, \tag{3.23}$$

where g_B is the $U(1)_B$ gauge coupling, m_{Z_B} the corresponding gauge boson mass, and $B_f = +1/3$ for quarks and 0 for leptons. Following the $B-L$ case, we adopt $m_{Z_B} = 10\,\text{TeV}$ so that the kinematic factor in Eq. (3.23) is effectively unity and the coupling reduces to $g_{fV} \approx \varepsilon\, g_B\, B_f$. The Z' decay width retains the form of Eq. (3.14), but with the sum running only over kinematically accessible quark–antiquark final states, since leptons are uncharged under $U(1)_B$.

For direct detection, the universal $B_q = 1/3$ quark charge again makes the spin-independent DM–nucleon cross section isospin-universal, taking the same form as Eq. (3.22) for the $B-L$ portal under the replacement $g_{B-L} \to g_B$:

$$\sigma_{\rm SI} \approx \frac{4\,\mu^2\,\alpha_X\,\varepsilon^2\,g_B^2}{m_{Z'}^4}\,. \tag{3.24}$$

As for the $B-L$ portal, only a single mediator contributes, so no blind spot arises.

3.4 `LiLjPortal`: $L_i - L_j$ Kinetic Mixing

The Z' mixes kinetically with the gauge boson of $U(1)_{L_i-L_j}$, acquiring purely vectorial couplings to leptons and their neutrinos of the two involved families [3]: flavour i carries charge $+1$ and flavour j carries charge -1, while quarks are uncharged. Three flavour combinations are supported: e–μ, μ–τ, and e–τ. The hidden-sector cross sections are identical to those of the `VectorPortal` (Eqs. (3.2)–(3.4)). Because quarks are uncharged, the Z' decays exclusively to leptons at tree level and the spin-independent DM–nucleon cross section vanishes, $\sigma_{\rm SI} = 0$. A nonzero $\sigma_{\rm SI}$ arises at loop level, but the resulting cross section is too small to produce meaningful direct-detection constraints unless the couplings are very large, so we neglect it.

3.5 `HiggsPortal`: Scalar Mixing

The DM particle X is a Majorana fermion ($g_X = 2$, no antiparticle doubling) coupled to a real scalar mediator ϕ ($g_Y = 1$) that mixes with the SM Higgs boson through the mixing angle ε, following the conventions of [3]. In the small-angle limit $\sin\varepsilon \approx \varepsilon$. Two couplings parametrise the hidden-sector Yukawa interactions:

$$\mathcal{L} \supset -\frac{\lambda_s}{2}\,\phi\,\bar{X}X - \frac{i\lambda_p}{2}\,\phi\,\bar{X}\gamma_5 X \tag{3.25}$$

where λ_s and λ_p are the scalar and pseudoscalar couplings respectively. We first describe the hidden-sector processes and then the connection to the SM through scalar mixing, which controls both the mediator decay width and the DM–nucleon scattering cross section.

3.5.1 Hidden-sector processes

The thermally averaged $2 \to 2$ annihilation cross section for $X\bar{X} \to \phi\phi$ is

$$\langle\sigma v\rangle_{X\bar{X}\to\phi\phi} = \left(a + \frac{6b}{x_X}\right), \tag{3.26}$$

where $x_X = m_X/T_h$, $r \equiv m_Y/m_X$, and

$$a = \frac{2\sqrt{1-r^2}\,\lambda_p^2\,\lambda_s^2}{m_X^2\,\pi\,(r^2-2)^2}, \tag{3.27}$$

$$b = \frac{1}{12\,m_X^2\,\pi\,\sqrt{1-r^2}\,(r^2-2)^4}\Big[-2(r^2-1)^3\,\lambda_p^4 + 3(r^6-8r^4+20r^2-12)\,\lambda_s^2\,\lambda_p^2 + 2(-2r^6+10r^4-17r^2+9)\,\lambda_s^4\Big] \tag{3.28}$$

are respectively the s-wave and p-wave coefficients. Note that the s-wave vanishes when either $\lambda_s = 0$ or $\lambda_p = 0$, so both couplings must be nonzero for a nonvanishing leading-order cross section. For the background evolution solver, only the s-wave piece $\langle\sigma v\rangle_{\text{s-wave}} = a$ is retained, since at late times ($x \gg 1$) the p-wave contribution is Boltzmann-suppressed and negligible. For $x_X < 1$ the p-wave term is also dropped, which is inconsequential since the hidden sector is still in full equilibrium in this regime.

The hierarchy among the three $3 \to 2$ cannibal processes mirrors that of the `VectorPortal`: $\phi\phi X \to \phi X$ dominates throughout freeze-out, $\phi X X \to X X$ is generically subdominant and tree-level, and $\phi\phi\phi \to \phi\phi$ is loop-suppressed. We compute Δ_2 and Δ_3 at tree level using `FeynCalc` [**? ? ? ?**], while Δ_1 is treated separately in the heavy-fermion limit (see below). The dominant channel takes the form

$$\langle\sigma v^2\rangle_{\phi\phi X\to\phi X} = \Delta_3(r,\rho)\,\frac{\lambda_s^3\,\lambda_p^3}{m_X^5} \tag{3.29}$$

where $r \equiv m_X/m_Y$, $\rho \equiv \lambda_p/\lambda_s$, and

$$\Delta_3(r,\rho) = \frac{9\sqrt{3}\,r^5\,\sqrt{3+8r+4r^2}}{64\pi\,(r+2)^3\,(2r+1)^4\,(2r^3+4r^2+r-1)^2} \times \Big[(1+2r)^5(3+2r) + (9+16r+180r^2+400r^3+288r^4+64r^5)(1+2r)^2\rho^2 + 3(-1+2r+2r^2)^2\rho^6 + (9+8r+140r^2+888r^3+2220r^4+3360r^5+3024r^6+1408r^7+256r^8)\rho^4\Big]. \tag{3.30}$$

The second channel, $\phi XX \to XX$, shares the same factorisation:

$$\langle \sigma v^2 \rangle_{\phi XX \to XX} = \Delta_2(r,\rho)\,\frac{\lambda_s^3\,\lambda_p^3}{m_X^5} \tag{3.31}$$

with

$$\begin{aligned}\Delta_2(r,\rho) = \frac{r^3\sqrt{1+4r}}{64\pi\,(1+2r)^3\,(-1+r+2r^2)^2} \times \Big[&(1+8r)(1+\rho^2)^3 + 4r^2(2+21\rho^2+20\rho^4+\rho^6) \\ &- 16r^3(4-7\rho^2-6\rho^4+5\rho^6) - 16r^4(7-6\rho^2-25\rho^4+4\rho^6) \\ &+ 128r^5(1+9\rho^2+14\rho^4+2\rho^6) + 256r^6(1+2\rho^2)^2\Big]\,. \end{aligned} \tag{3.32}$$

This channel is generally subdominant but is included for completeness. The third channel, $\phi\phi\phi \to \phi\phi$, factorises with a different coupling dependence:

$$\langle \sigma v^2 \rangle_{\phi\phi\phi \to \phi\phi} = \Delta_1(r,\rho)\,\frac{\lambda_s^5\,\lambda_p^5}{m_X^5} \tag{3.33}$$

with

$$\Delta_1(r,\rho) = \frac{\sqrt{5}}{3\pi^5}\,\frac{r^3\,(3+10\rho^2+15\rho^4)^2}{\rho^3}\,. \tag{3.34}$$

Unlike the tree-level channels above, $\phi\phi\phi \to \phi\phi$ proceeds through a one-loop pentagon diagram (a closed X loop with five external ϕ legs) and the expression above is derived in the heavy-fermion limit $r \gg 1$, where all external momenta are negligible compared to the loop momentum. This is self-consistent as the $\phi\phi\phi \to \phi\phi$ channel can only become competitive with $\phi\phi X \to \phi X$ precisely in this regime, since at low r the dark matter X is comparably abundant to ϕ and the tree-level process wins without any loop penalty. At large r, however, X is exponentially Boltzmann-suppressed in the initial state, and the cannibal process — which only requires X as a virtual particle inside the loop — sidesteps this suppression, compensating the $\sim\lambda^4/(16\pi^2)^2$ loop cost.

3.5.2 Connection to the Standard Model

Mediator decay width. The mediator ϕ decays to SM particles through its mixing with the Higgs boson. All partial widths are proportional to $\sin^2\varepsilon \approx \varepsilon^2$, and we define the normalised total width $\Gamma_{\rm norm}(m_\phi) \equiv \Gamma_{\rm tot}/\sin^2\varepsilon$ so that the physical width is

$$\Gamma(\phi \to \text{SM}) = \varepsilon^2\,\Gamma_{\rm norm}(m_\phi)\,. \tag{3.35}$$

The SM-like contribution to $\Gamma_{\rm norm}$ is computed using a hybrid approach across four mass regimes:

- $m_\phi < 5\,\text{GeV}$: the `scalar_portal` package [4], which uses a dispersive analysis below 2 GeV (leptonic decays, hadronic channels via form factors, multi-meson final states) and perturbative QCD with running quark masses ($N_f = 4$) above 2 GeV.
- $5\,\text{GeV} \le m_\phi \le 500\,\text{GeV}$: the `HDECAY` Fortran program [5, 6], which includes higher-order QCD corrections, off-shell decays to WW^*/ZZ^*, and loop-induced channels (gg, $\gamma\gamma$).

- $m_\phi > 500\,\mathrm{GeV}$: tree-level decay widths with exact kinematics, since `HDECAY` is not reliable in this regime due to un-resummed Sudakov logarithms.

If $m_\phi > 2m_h$, an additional contribution from $\phi \to hh$ arises in the singlet extension:

$$\frac{\Gamma(\phi \to hh)}{\sin^2\varepsilon} = \frac{(2m_\phi^2 + m_h^2)^2}{128\pi\, m_\phi\, v_H^2} \sqrt{1 - \frac{4m_h^2}{m_\phi^2}} \tag{3.36}$$

with $v_H = 246.22\,\mathrm{GeV}$ the Higgs vacuum expectation value. This channel can be switched off by setting `hh_coupling='zero'`.

Direct-detection cross section. The spin-independent DM–nucleon scattering cross section proceeds via t-channel exchange of the two mixed scalars h and ϕ. After electroweak symmetry breaking, the mass mixing $\phi \to \phi - \varepsilon\, h$ generates an effective ϕ–nucleon coupling proportional to $\varepsilon\,\lambda_s$. In the small-mixing limit, the h- and ϕ-exchange amplitudes add coherently with opposite sign (since the mixing rotates them in opposite directions), giving a per-nucleon amplitude proportional to the propagator difference $1/m_Y^2 - 1/m_h^2$. The resulting cross section is

$$\sigma_{\mathrm{SI}} = \frac{4\mu^2}{\pi}\, a_N^2\, \hbar^2 c^2\,, \qquad a_N = \frac{\lambda_s\, \varepsilon\, f_N\, m_N}{v_H} \left(\frac{1}{m_Y^2} - \frac{1}{m_h^2} \right) \tag{3.37}$$

where $f_N \simeq 0.30$ is the Higgs–nucleon form factor, m_h the SM Higgs mass, and μ the DM–nucleon reduced mass. The factor of 4 reflects the Majorana nature of X (identical-particle combinatorics). This result is isospin-universal and independent of the target nucleus composition, since the coupling to nucleons proceeds entirely through the Higgs–nucleon vertex. The numerical estimate in [1] (their Eq. 67) is consistent with this expression in the limit $m_Y \ll m_h$. The propagator difference vanishes at $m_Y = m_h$, where the two amplitudes cancel exactly and $\sigma_{\mathrm{SI}} = 0$. At this blind spot the required $\varepsilon \to \infty$, making the model undetectable by direct-detection experiments regardless of the hidden-sector couplings.

The default parameter values used in the `HiggsPortal` class are summarised in Table 2.

Parameter	Symbol	Default
DM mass	m_X	$100\,\mathrm{GeV}$
Mediator mass	m_Y	$30\,\mathrm{GeV}$
Dark couplings	$\lambda_s = \lambda_p$	10^{-2}
DM DOF	g_X	2 (Majorana fermion)
Mediator DOF	g_Y	1 (real scalar)
Cannibal rescaling	$\Delta_1, \Delta_2, \Delta_3$	exact tree-level
Initial temperature ratio	ξ_{ini}	1.0
$\phi \to hh$ coupling	`hh_coupling`	`'singlet'`

Table 2: Default parameters for the `HiggsPortal` model.

4 Evolution of the Hidden Sector

The cosmological evolution of the hidden sector can be naturally divided into two stages, each amenable to a distinct formulation. The first stage tracks the coupled evolution of the DM particle X, the mediator Y, and the hidden-sector temperature as the hidden sector departs from internal chemical equilibrium. The key processes during this phase are the $2 \to 2$ annihilation $X\bar{X} \to YY$, which sets the DM relic abundance, and the $3 \to 2$ cannibal reactions, which maintain the mediator in chemical equilibrium until they too freeze out. Throughout this work, the portal coupling ε is assumed to be sufficiently small that the hidden sector remains thermally decoupled from the Standard Model bath, evolving independently at its own temperature $T_h = \xi\, T_{\rm SM}$. In this regime, the mediator decay width generically satisfies $\Gamma_Y \ll H$ during the entire freeze-out process, so that it can be safely neglected and the dynamics is governed solely by the hidden-sector couplings. The evolution during this stage is then captured by a set of coupled Boltzmann equations, which we present in Sec. 4.1. We first write these equations in their standard, physically transparent form and then show how they can be recast in terms of chemical-potential variables and the hidden to SM temperature ratio ξ, a formulation that is better suited for numerical integration.

The second stage begins once the Boltzmann system has converged and the comoving abundances of X and Y are effectively frozen. The relevant dynamics is now the decay of the long-lived mediator into SM particles. As Y depletes, it injects entropy into the visible sector and dilutes the dark-matter yield; it is through this mechanism that the portal coupling ε ultimately controls the predicted relic abundance. This late-time background evolution is described in Sec. 4.2.

4.1 Boltzmann Equations

The starting point is the standard set of Boltzmann equations for the number densities n_X and n_Y and the total hidden-sector energy density $\rho_h = \rho_X + \rho_Y$. In a Friedmann–Robertson–Walker background these read:

$$\dot{n}_X + 3Hn_X = -\langle\sigma v\rangle_X \left(n_X^2 - (n_X^{\rm eq})^2 \, \frac{n_Y^2}{(n_Y^{\rm eq})^2} \right) \tag{4.1}$$

$$\dot{n}_Y + 3Hn_Y = \langle\sigma v\rangle_X \left(n_X^2 - (n_X^{\rm eq})^2 \, \frac{n_Y^2}{(n_Y^{\rm eq})^2} \right) - \Gamma_{\rm can}\,(n_Y - n_Y^{\rm eq}) \tag{4.2}$$

$$\dot{\rho}_h + 3H(\rho_h + P_h) = 0 \tag{4.3}$$

where a dot corresponds to a derivative with respect to cosmic time, H is the total Hubble rate including the hidden-sector contribution to the energy density, Eq. (2.5), $\langle\sigma v\rangle_X$ is the thermally averaged annihilation cross section, and $\Gamma_{\rm can}$ collects all $3 \to 2$ number-changing processes, namely:

$$\Gamma_{\rm can} = \langle\sigma v^2\rangle_{YYY\to YY}\, n_Y^2 \; + \langle\sigma v^2\rangle_{YYX\to YX}\, n_Y\, n_X + \; \langle\sigma v^2\rangle_{YXX\to XX}\, n_X^2 \,. \tag{4.4}$$

The first equation governs the DM abundance, with the collision term vanishing when $n_X/n_Y = n_X^{\rm eq}/n_Y^{\rm eq}$ to enforce detailed balance. The second equation tracks the mediator, which is sourced by annihilation and depleted by cannibal reactions. The third equation expresses energy conservation within the hidden sector and determines the evolution of the hidden-sector temperature T_h. Under the Maxwell–Boltzmann statistics adopted throughout this work, the energy density and pressure

of each species are fully specified by its number density and temperature through $\rho_i = B_1^i\, n_i$ $\rho_i + P_i = B_2^i\, n_i$, Eqs. (2.8) and (2.9), so that Eq. (4.3) can be rewritten as an evolution equation for the hidden-sector temperature T_h given n_X and n_Y.

Finally, the equations above are written for self-conjugate X (Majorana fermion), following the conventions of [1]. When X is not self-conjugate, i.e. a Dirac fermion, n_X counts both particles and antiparticles and the cross sections $\langle\sigma v\rangle_X$ and $\langle\sigma v^2\rangle_{YXX\to XX}$ are each replaced by half their value to avoid double-counting identical pairs in the initial state.

4.1.1 Reformulation in terms of chemical potentials and ξ

Rather than evolving the number densities directly, it is advantageous to reformulate the system in terms of the dimensionless chemical potentials $\bar{\mu}_i \equiv \mu_i/T_h$, defined through

$$Y_i = Y_i^{\rm eq}\, e^{\bar{\mu}_i}\,, \qquad i \in \{X, Y\}\,, \tag{4.5}$$

where $Y_i = n_i/s$ is the comoving yield, s is the SM entropy density and $Y_i^{\rm eq} = n_{{\rm MB},\,i}^{\rm eq}/s$ its equilibrium value. This parametrization has two key benefits. First, in the early universe when both species are in chemical equilibrium, $\bar{\mu}_i = 0$ exactly, avoiding the need to track the cancellation between Y_i and $Y_i^{\rm eq}$ — a severe numerical challenge when both are large and nearly equal. Second, during freeze-out the dimensionless chemical potential grows linearly while the yield departs exponentially, so $\bar{\mu}_i$ is a much more slowly varying quantity and hence better suited as an ordinary differential equation (ODE) variable.

We take as independent variable $x \equiv m_X/T_{\rm SM}$ and promote the dark-to-SM temperature ratio $\ln\xi$ to a dynamical degree of freedom, obtaining a coupled system of three ODEs:

$$\frac{d\bar{\mu}_X}{dx} = A + B\,\frac{d\ln\xi}{dx} \tag{4.6}$$

$$\frac{d\bar{\mu}_Y}{dx} = C + D\,\frac{d\ln\xi}{dx} \tag{4.7}$$

$$\frac{d\ln\xi}{dx} = \frac{E + F\,A + G\,C}{1 - F\,B - G\,D} \tag{4.8}$$

where the six coefficient functions A–G depend on $(x,\, \xi,\, \bar{\mu}_X,\, \bar{\mu}_Y)$ and are constructed from the thermally averaged cross sections, the equilibrium thermodynamic quantities $(B_1,\, B_2,\, \lambda)$, and the SM entropy-conservation factor $\tilde{g}$ as defined in Secs. 2 and 3. The explicit forms for these coefficients are given in Eqs. (4.9)–(4.14) below. The full derivation of this system from the number-density Boltzmann equations (4.1)–(4.3), including the extension to nonzero mediator decay width, is given in Appendix C.

Dark-matter coefficients.

$$A = -\frac{s\,\tilde{g}}{H_{\rm tot}\,x}\,\langle\sigma v\rangle_X Y_X^{\rm eq} e^{\bar{\mu}_X}\left[1 - e^{2(\bar{\mu}_Y - \bar{\mu}_X)}\right] + \frac{\lambda_X - 3\tilde{g}}{x}\,, \tag{4.9}$$

$$B = -\lambda_X\,. \tag{4.10}$$

The first term in A is the collision integral for $X\bar{X} \leftrightarrow YY$, written in a form that makes the approach to chemical equilibrium ($\bar{\mu}_X = \bar{\mu}_Y$) manifest. The second term encodes the redshifting of the equilibrium yield, with $\lambda_X \equiv d\ln n_X^{\rm eq}/d\ln T_h$, Eq. (2.7), capturing the temperature sensitivity of the equilibrium density and $\tilde{g}$, Eq. (2.4), which accounts for the running of the SM entropy degrees of freedom.

Mediator coefficients.

$$C = \frac{\tilde{g}}{H_{\rm tot}\, x}\left\{ s\,\langle\sigma v\rangle_X\, \frac{(Y_X^{\rm eq})^2}{Y_Y^{\rm eq}} e^{2\bar{\mu}_X - \bar{\mu}_Y}\, \left[1 - e^{2(\bar{\mu}_Y - \bar{\mu}_X)}\right] - \Gamma_{\rm can}(1 - e^{-\bar{\mu}_Y}) \right\} + \frac{\lambda_Y - 3\tilde{g}}{x}\,, \tag{4.11}$$

$$D = -\lambda_Y\,, \tag{4.12}$$

The first contribution in C is the annihilation source, equal and opposite to the corresponding term in A. The second involves the total cannibal reaction rate $\Gamma_{\rm can}$ defined in Eq. (4.4), multiplied by the factor $(1 - e^{-\bar{\mu}_Y})$ which vanishes identically when Y is in chemical equilibrium ($\bar{\mu}_Y = 0$) and acts as a restoring force that drives $\bar{\mu}_Y$ back toward zero whenever cannibal reactions are fast. The last term plays the same role as in A, accounting for the redshifting of $Y_Y^{\rm eq}$.

Temperature-equation coefficients. These arise from projecting the energy-conservation equation onto $d\ln\xi/dx$:

$$E = \frac{1}{x}\left(1 - 3\tilde{g}\,\frac{n_X\, B_2^X + n_Y\, B_2^Y}{\tilde{D}}\right), \tag{4.13}$$

$$F = -\frac{B_1^X\, n_X}{\tilde{D}}\,, \qquad G = -\frac{B_1^Y\, n_Y}{\tilde{D}}\,, \tag{4.14}$$

with the energy denominator,

$$\tilde{D} = T_h\left(n_X\,\frac{dB_1^X}{dT_h} + n_Y\,\frac{dB_1^Y}{dT_h}\right) + B_1^X\, n_X\,\lambda_X + B_1^Y\, n_Y\,\lambda_Y\,. \tag{4.15}$$

Here, the coefficient E captures the adiabatic cooling of the hidden sector due to Hubble expansion, while F and G encode the feedback of changes in n_X and n_Y on the hidden-sector temperature. All three are controlled by the denominator $\tilde{D}$, which measures the total heat capacity of the dark plasma. The thermodynamic quantities B_1, B_2, and λ used here are defined in Sec. 2.2.

4.1.2 Solver Implementation

The system of Eqs. (4.6)–(4.8) is solved by the function `solve_boltzmann_chempot_3phase`, which employs a three-phase strategy that exploits the natural hierarchy of timescales in the hidden sector. The DM is the first species to depart from chemical equilibrium, at a time when cannibal processes are still fast enough to hold the mediator at its equilibrium abundance. Only later do the cannibal reactions themselves become inefficient, and the mediator too falls out of equilibrium, entering the full freeze-out regime. Solving the full three-variable system from the outset would introduce unnecessary stiffness in the early evolution, where the mediator equation is effectively algebraic. The three-phase decomposition avoids this by progressively activating degrees of freedom as each process falls out of equilibrium.

In Phase 1, both species are in chemical equilibrium, i.e. $\bar{\mu}_X = \bar{\mu}_Y = 0$, and only the hidden-sector temperature evolves. The system reduces to a single ODE for $\ln\xi$, obtained from Eq. (4.8) with all equilibrium densities evaluated at T_h. In formulas:

$$\left.\frac{d\ln\xi}{dx}\right|_{\rm eq} = \frac{1}{x}\left(1 - 3\tilde{g}\,\frac{n_Y^{\rm eq}\, B_2^Y + n_X^{\rm eq}\, B_2^X}{T_h\left(B_1^X\,\frac{dn_X^{\rm eq}}{dT_h} + \frac{dB_1^X}{dT_h}\,n_X^{\rm eq} + B_1^Y\,\frac{dn_Y^{\rm eq}}{dT_h} + \frac{dB_1^Y}{dT_h}\,n_Y^{\rm eq}\right)}\right) \tag{4.16}$$

The solver exits this phase when the dimensionless annihilation rate $\Gamma_{\rm ann}/H \equiv (s/H_{\rm tot})\,\langle\sigma v\rangle_X Y_X$ drops below a configurable threshold safely set at 5×10^9 by default (the parameter `Gamma_over_H_ann` in the code).

In Phase 1.5, the DM has begun to depart from equilibrium but cannibal reactions remain fast enough to enforce $\bar{\mu}_Y = 0$. Setting $\bar{\mu}_Y = 0$ and $d\bar{\mu}_Y/dx = 0$ in Eqs. (4.6)–(4.8) eliminates the mediator equation entirely and simplifies the coefficients A and C, since all factors of Y_Y reduce to $Y_Y^{\rm eq}$. The system then reduces to two ODEs for $(\ln\xi,\, \bar{\mu}_X)$, with the mediator abundance fixed at its equilibrium value. This intermediate phase is the key ingredient that resolves the stiffness near the annihilation freeze-out transition. It ends when $\Gamma_{\rm can}/H$ (see Eq. (4.4)) drops below a configurable threshold, set to unity by default (the parameter `Gamma_over_H_can` in the code), at which point the cannibal processes can no longer maintain the mediator in chemical equilibrium.

In Phase 2, all three variables $(\ln\xi,\, \bar{\mu}_X,\, \bar{\mu}_Y)$ evolve via the full system of Eqs. (4.6)–(4.8) with the coefficients given in Eqs. (4.9)–(4.15). For large mass ratios $r \equiv m_X/m_Y$, the mediator freezes out much later than the DM, and the integration range is extended accordingly to $x_{\rm max} = \max(10^5,\, r^2 \times 10^4)$.

The initial conditions are set at $x_0 = x_{\rm min}$ set by default to 10^{-3}, deep in the equilibrium regime. The chemical potentials start at zero, $\bar{\mu}_X(x_0) = \bar{\mu}_Y(x_0) = 0$, with the initial hidden to SM temperature ratio determined by entropy conservation from a high reference temperature $T_{\rm ini} = 10^{14}$ GeV by default. In formulas we get:

$$\ln\xi(x_0) = \frac{1}{3}\ln\left(\frac{g_{*S}(m_X/x_0)}{g_{*S}(T_{\rm ini})}\right) + \ln\xi_{\rm ini}\,, \tag{4.17}$$

where $\xi_{\rm ini}$ is the corresponding asymptotic hidden to SM temperature ratio at the reference temperature, a free parameter of the model. It is important to emphasize that the precise value of $T_{\rm ini}$ is not relevant, provided it lies well above all mass scales in the problem and above the highest threshold at which $g_{\star S}$ changes, so that $g_{\star S}(T_{\rm ini})$ has reached its asymptotic value.

The integration is terminated once the system has converged. Convergence is checked for $x > 2\,x_{\rm FO}$, where $x_{\rm FO}$ is the value of x at which $\bar{\mu}_X$ first exceeds 0.05, by requiring that the hidden-sector temperature tracks the non-relativistic scaling expected when all species have frozen out, namely that:

$$\left.\frac{d\ln\xi}{d\ln x}\right|_{\rm NR} = 1 - 2\tilde{g}\,. \tag{4.18}$$

Specifically, the solver terminates once $|d\ln\xi/d\ln x - (1-2\tilde{g})|$ falls below a configurable tolerance, set to 10^{-2} by default (the parameter `convergence_threshold` in the code).

4.2 Late-time background evolution and mediator decay

Once the Boltzmann system has converged, the comoving yields of X and Y are effectively frozen and the hidden-sector temperature has settled onto the non-relativistic scaling of Eq. (4.18). The subsequent evolution is governed by the decay of the mediator into SM particles, $Y \to$ SM, with a decay width $\Gamma_Y \propto \varepsilon^2$ whose explicit form depends on the portal model (see Sec. 3) and is now the only relevant timescale. This stage is handled by a separate solver, `solve_background`, which evolves the system in the number of e-folds $N = \ln(a/a_0)$ using three log-scaled variables, namely

$y_0 = \ln(T/T_0)$, $y_1 = \ln(\rho_Y/\rho_{Y,0})$, and $y_2 = \ln(\rho_X/\rho_{X,0})$, where the subscript "0" denotes values at the start of the background evolution. The evolution equations read

$$\frac{d\ln T}{dN} = -\frac{1}{\tilde{g}}\left(1 - \frac{\rho_Y\,\Gamma_Y}{3H_{\rm tot}\,s\,T}\right) \tag{4.19}$$

$$\frac{d\ln \rho_Y}{dN} = -\left(3 + \frac{\Gamma_Y}{H_{\rm tot}}\right) \tag{4.20}$$

$$\frac{d\ln \rho_X}{dN} = -3 - \frac{\langle\sigma v\rangle_{\text{s-wave}}\,\rho_X}{m_X\,H_{\rm tot}}\left[1 - \left(\frac{n_Y\,n_X^{\rm eq}}{n_X\,n_Y^{\rm eq}}\right)^2\right] \tag{4.21}$$

with the total Hubble rate $H_{\rm tot}$ is evaluated at each step from Eq. (2.5). The first equation describes the evolution of the SM temperature, which now receives an entropy injection term from the decay $Y \to$ SM that slows the usual $T \propto a^{-1}$ redshifting. The second combines Hubble dilution of the mediator energy density with its depletion by decay, while the third tracks the DM energy density retaining the residual contributions from both annihilations and inverse annihilations $YY \to X\bar{X}$ with equilibrium densities evaluated at the redshifted hidden-sector temperature $T_h \propto a^{-1}$. Both are typically negligible at this stage but are retained for completeness, and the inverse rate is dynamically relevant in the joint workflow below where ξ remains elevated at the start of the background phase.

The key observable is the total entropy injected into the SM bath, quantified by the ratio $S_{\rm ratio} = S_{\rm final}/S_0$ of the final to initial comoving entropy $S = s\,a^3$. When the mediator decays promptly ($\Gamma_Y \gg H$), the entropy injection is negligible and $S_{\rm ratio} \approx 1$. Instead, for late decays $S_{\rm ratio}$ can be $\gg 1$ and the DM yield is diluted accordingly to $Y_X^{\rm final} = \rho_X^{\rm final}/(m_X s_{\rm final}) \approx Y_{X,0}/S_{\rm ratio}$. The evolution is terminated when $\rho_Y/\rho_{\rm tot} < 10^{-12}$, ensuring that the mediator has fully decayed and the entire entropy injection has been accounted for.

Decoupling the portal coupling from the Boltzmann evolution. A key design choice of the solver is to treat the mediator as stable during the Boltzmann evolution of Sec. 4.1.2, neglecting the portal coupling entirely. The Boltzmann system is numerically expensive due to its stiffness and wide dynamic range, so it is highly advantageous to solve it once for a given set of hidden-sector couplings and masses, and then scan over ε using only the fast background evolution to find the value that reproduces the observed relic abundance. For $\varepsilon \lesssim 10^{-10}$, where $\Gamma_Y \ll H$ throughout freeze-out, this approximation is excellent. An optional post-hoc correction (`correct_Y_decay`) can further improve its accuracy by retroactively stepping through the Boltzmann trajectory, computing the cumulative decay and entropy injection that was neglected, and adjusting the initial conditions for the background evolution according to

$$Y_{Y,0} \to Y_{Y,0}\,f_{\rm survive}\,, \qquad T_0 \to T_0\left(\frac{S_{\rm corr}}{S_0}\right)^{1/3}, \qquad Y_{X,0} \to \frac{Y_{X,0}}{S_{\rm corr}/S_0}\,. \tag{4.22}$$

Specifically, the net energy injected into the SM bath is accumulated along the stored trajectory, and the corresponding entropy ratio follows in closed form,

$$I \equiv \int dt\,\frac{\Gamma_Y\left(\rho_Y - \rho_Y^{\rm eq,SM}\right)}{s\,T}\,, \qquad \frac{S_{\rm corr}}{S_0} = \left(1 + \frac{4}{3}\,I\right)^{3/4}, \tag{4.23}$$

where ρ_Y is the physical mediator energy density, $\rho_Y^{\rm eq,SM}$ its equilibrium value at the SM temperature, which accounts for inverse decays, and s and T are those of the trajectory without injection. The closed form follows from the instantaneous relation $dS/S = \Gamma_Y\,\rho_Y\,dt/(s\,T)$ once the reheating of the SM bath is taken into account: at fixed g_{*S} the comoving entropy scales as $S \propto (aT)^3$, so that $S^{1/3}\,dS$ is set by the trajectory alone and integrates to $(S_{\rm corr}/S_0)^{4/3} = 1+4I/3$. Its first-order expansion, $S_{\rm corr}/S_0 \simeq e^I$, is accurate only while the injected energy remains small compared with that of the SM bath. For hot hidden sectors near threshold it overshoots by orders of magnitude, yielding entropy ratios of 50–1700 at $\xi_{\rm ini} = 2.5$ and $m_Y/m_X \simeq 0.87$, where energy conservation allows roughly 5. Since the injection heats only the SM bath, the hidden-sector temperature at the handoff is left unchanged, and the temperature ratio is reset to $\xi_0 = T_{h,0}/T_0$ with the reheated T_0 of Eq. (4.22). Retaining the uncorrected ξ_0 would instead overstate T_h, spuriously reviving the inverse annihilations $YY \to X\bar{X}$ in Eq. (4.21). The same factor renormalizes the handoff state of the entropy-injection phase of the joint solver (Phase 3, `enable_phase3`), which in addition rejects any trajectory with $S_{\rm corr}/S_0 > 2$, requiring an earlier handoff rather than a large post-hoc correction.

This is a precision refinement within the secluded workflow. It does not address the regime where Γ_Y becomes comparable to H during freeze-out itself, where the persistent depletion of Y by decay limits the $YY \to X\bar{X}$ backreaction and alters the freeze-out dynamics in a way that no adjustment of the handoff conditions can capture.

Joint evolution beyond the secluded approximation. To extend the framework into this regime, the code provides an alternative solver, `solve_boltzmann_joint`, which evolves the hidden sector together with the portal-induced $Y \leftrightarrow$ SM exchanges throughout freeze-out. This modifies two of the seven coefficient functions of Eqs. (4.6)–(4.8), those governing the mediator number density and the hidden-sector temperature,

$$C \to C - \frac{\tilde{g}}{H_{\rm tot}\,x}\,\Gamma_Y\left(1 - \frac{n_Y^{\rm eq,SM}}{n_Y}\right), \tag{4.24}$$

$$E \to E - \frac{\tilde{g}}{H_{\rm tot}\,x}\,\frac{\Gamma_Y\,(B_1^Y\,n_Y - B_1^{Y,\,\rm SM}\,n_Y^{\rm eq,SM})}{\tilde{D}}, \tag{4.25}$$

where $n_Y^{\rm eq,SM}$ is the mediator equilibrium number density evaluated at the SM temperature, $B_1^{Y,\,\rm SM}$ the corresponding energy-per-particle coefficient, and $\tilde{D}$ the energy denominator of Eq. (4.15). The modification to C combines the forward decay $Y \to$ SM that depletes the mediator yield with the inverse decay SM $\to Y$ that repopulates it, while the modification to E captures the corresponding net energy flow. When both rates exceed Hubble, $n_Y \to n_Y^{\rm eq,SM}$ and the hidden sector thermalises with the SM bath, recovering the standard WIMP picture.

The joint solver subdivides the intermediate phase of Sec. 4.1.2 into two branches, entered automatically depending on the local rates. The *dark-locked* branch (Phase 1.5-D) is the direct extension of the secluded Phase 1.5, where cannibal reactions pin $\bar{\mu}_Y = 0$, ξ remains a dynamical degree of freedom, and Eqs. (4.24)–(4.25) provide the only source and sink to the SM bath. The *SM-locked* branch (Phase 1.5-S) is entered when the per-particle inverse-decay rate satisfies $\Gamma_Y\,n_Y^{\rm eq,SM}/(H\,n_Y) \geq 1$, at which point Y has reached detailed balance against the SM bath and the temperature ratio is pinned to $\xi = 1$. The only remaining degree of freedom is then $\bar{\mu}_X$,

and the system reduces to a single ODE that is equivalent to the corresponding equation in the comoving yield Y_X, which the dedicated solver `solve_boltzmann_thermSM` (see Appendix B.1 for details) can also integrate directly when one knows *a priori* that the system is fully thermalised.

The joint solver therefore covers the full range of portal couplings. In the deeply secluded regime it recovers the primary solver at the sub-percent level. For intermediate values of ε, only the forward decay is efficient and the freeze-out is affected by the persistent depletion of Y. At the upper end it reaches the thermalisation floor of Sec. 5.4. In practice the dimensionless ratio Γ_Y/H evaluated at the handoff between the Boltzmann and background phases (returned as `GoH_handoff`) determines which workflow is needed, and when it drops well below unity the cheaper `solve_boltzmann_chempot_3phase` is preferred.

5 Mapping the Portal Coupling: Predictions and Constraints

The framework developed in Sec. 4 provides, for any choice of the hidden-sector parameters — masses m_X and m_Y, and the hidden-sector couplings (e.g. α_X for the vector portals, or λ_s, λ_p for the Higgs portal) — the late-time DM yield $Y_X^{\rm final}$ and the frozen mediator abundance $Y_Y^{\rm frozen}$ as functions of the portal coupling ε. Because ε enters only through the mediator decay width, it controls the late-time entropy injection that dilutes the relic abundance but does not affect the hidden-sector freeze-out itself, which is governed entirely by the hidden-sector couplings. This separation makes ε the natural variable through which to express both predictions and observational constraints.

In this section we describe how the toolkit uses this dependence to determine three quantities in the (m_X, ε) plane, at fixed hidden-sector couplings and mass ratio $r = m_X/m_Y$:

- the value $\varepsilon_{\rm relic}(m_X)$ that reproduces the observed DM relic abundance $\Omega_c h^2 = 0.12$ — our *prediction line* (Sec. 5.1);
- the lower bound $\varepsilon_{\rm BBN}(m_X)$ from Big Bang Nucleosynthesis (BBN) hadronic injection by late mediator decays (Sec. 5.2);
- the upper bound $\varepsilon_{\rm DD}(m_X)$ from spin-independent DM–nucleon scattering experiments (Sec. 5.3).

The intersection of the relic line with the excluded regions determines the viable parameter space of the model. A separate requirement — that the hidden sector remain thermally decoupled from the SM bath — sets a thermalization floor on ε that we discuss in Sec. 5.4.

5.1 Dark Matter Relic Abundance

For a given choice of hidden-sector parameters, the value of the portal coupling that reproduces the observed DM relic abundance is determined by the function `find_epsilon_DMRelic`. The procedure runs the Boltzmann solver of Sec. 4.1.2 once to obtain the frozen-out yields, and then performs a one-dimensional root search on the background evolution of Sec. 4.2 for the value of ε that satisfies

$$Y_X^{\rm final}(\varepsilon) = Y_{\rm target}\,, \qquad Y_{\rm target} \equiv \frac{\Omega_c h^2\,\rho_c}{s_0\,m_X}\,, \tag{5.1}$$

with $\Omega_c h^2 = 0.12$, $\rho_c = 1.05375 \times 10^{-5}\, h^2\,\mathrm{GeV/cm^3}$, and $s_0 = 2891\,\mathrm{cm^{-3}}$. The root search is performed on $\log_{10}\varepsilon$ via Brent's method (`scipy.optimize.brentq`) over a default range $\log_{10}\varepsilon \in [-20, -6]$.

At fixed masses and hidden-sector couplings, the freeze-out fixes both Y_X^{frozen} and Y_Y^{frozen} at the end of the Boltzmann phase. The role of ε is then to set the lifetime of Y and hence the magnitude of the entropy injected into the visible sector once Y decays. Larger ε corresponds to a shorter lifetime, less entropy injection, and a less diluted relic. There is therefore a unique $\varepsilon = \varepsilon_{\mathrm{relic}}(m_X)$ for which the final yield matches Y_{target}, provided Y_X^{frozen} is itself *at least* as large as Y_{target} — otherwise no amount of entropy dilution can recover the observed abundance.

Inverted workflow at large portal coupling. The procedure above breaks down when ε approaches the thermalisation floor, where the secluded picture underlying `find_epsilon_DMRelic` is no longer accurate and the root search may fail to bracket a solution. The natural approach in this regime is to fix ε and search instead for the DM mass that reproduces the observed relic abundance. This is implemented by the function `find_mX_DMRelic_joint`, which uses the joint solver of Sec. 4.2 for the Boltzmann phase (see also Appendix B.1). The mass scale is first estimated from the standard freeze-out relation $m_X^{\mathrm{est}} = [\langle\sigma v\rangle_0\,(mY)_{\mathrm{relic}}\,M_{\mathrm{Pl}}]^{1/2}$, with $\langle\sigma v\rangle_0$ the model's s-wave thermally averaged cross section evaluated at a reference DM mass of 1 GeV and $(mY)_{\mathrm{relic}} \equiv \Omega_c h^2\,\rho_c/s_0$. The joint workflow is then evaluated on a bracket of n_{refine} mass points (default 5) log-spaced around m_X^{est}, and $m_X(\varepsilon)$ is obtained by linear interpolation in $[\log_{10} m_X, \log_{10}(\Omega_X h^2)]$. An optional Brent polish on the converged bracket is available at the cost of additional joint solves.

5.2 Constraints from Big Bang Nucleosynthesis

The second quantity of interest is the lower bound on the portal coupling set by BBN. In the hidden-sector scenario, the hidden-sector freeze-out leaves behind a sizeable comoving abundance of mediators Y which, for sufficiently small ε, can survive long enough to decay during or after BBN and inject hadronic energy into the primordial plasma. If this injection is too large, it spoils the successful predictions of the light-element abundances, and the corresponding region of parameter space is excluded. The code adopts the constraints derived in [7], which are tabulated as an upper bound on the energy yield per decay $\zeta \equiv m_Y\,Y_Y$ (in GeV) as a function of the mediator lifetime $\tau_Y \equiv 1/\Gamma_Y$ (in s) for several reference masses and two representative hadronic injection channels, $u\bar{u}$ and $b\bar{b}$. A two-dimensional log–log interpolation (`scipy.interpolate.RegularGridInterpolator`) is built in the $(\log_{10}(m_Y/\mathrm{TeV}), \log_{10}\tau_Y)$ plane over six reference masses (30 GeV, 100 GeV, 1 TeV, 10 TeV, 100 TeV, 1000 TeV). Outside the tabulated mass range the bound is extended via a power-law extrapolation:

$$\zeta_{\max}(m_Y) = \zeta_{\max}(m_{\mathrm{anchor}})\,(m_Y/m_{\mathrm{anchor}})^{\alpha} \tag{5.2}$$

where the slope α is extracted numerically as the local power-law index $\Delta(\log_{10}\zeta_{\max})/\Delta(\log_{10} m_Y)$ between adjacent tabulated masses at fixed τ_Y, evaluated at the edge of the grid closest to the extrapolation region. For $m_Y > 1000\,\mathrm{TeV}$ (anchored to the highest tabulated point), both channels converge to $\alpha_{\mathrm{high}} \simeq 0.70$. This value follows from the QCD fragmentation multiplicity scaling $N_{\mathrm{had}} \sim m_Y^{0.3}$ [7], which implies that a heavier particle produces more hadrons per decay, so fewer

decays are needed to inject the same hadronic energy, and the upper bound on the total energy yield relaxes as $\zeta_{\max} \propto m_Y^{1-0.3}$. For $m_Y < 30\,\mathrm{GeV}$ (anchored to the lowest tabulated point), the slopes are reduced to $\alpha_{\mathrm{low}} = 0.50$ for $u\bar{u}$ and 0.34 for $b\bar{b}$. The departure from the asymptotic value of 0.7 arises because at lower masses the energy per nucleon in the shower, $E_N \sim m_Y/N_{\mathrm{had}} \sim m_Y^{0.7}$, drops into a regime where the hadrodissociation cross-sections decrease with energy, so that each individual hadron becomes less effective at disrupting light elements and the bound tightens more slowly with mass than the multiplicity argument alone would predict. The particularly low $b\bar{b}$ slope reflects an additional suppression from the proximity to the b-quark threshold ($2m_b \approx 8.4\,\mathrm{GeV}$), which sharply reduces shower multiplicity relative to $u\bar{u}$.

Given the interpolated bound, for a given (m_X, m_Y) and fixed hidden-sector couplings, the minimum portal coupling compatible with BBN, $\varepsilon_{\mathrm{BBN}}(m_X)$, is obtained by a one-dimensional root search on $\log_{10}\varepsilon$ of the condition

$$m_Y\, Y_Y^{\mathrm{frozen}} = \zeta_{\max}(m_Y,\, \tau_Y(\varepsilon)) \ , \tag{5.3}$$

with Y_Y^{frozen} read off from the Boltzmann solver output of Sec. 4.1.2.

Across essentially the entire parameter space considered in this work, namely for $m_X \gtrsim m_X^\star$, the mediator abundance left over by the hidden-sector freeze-out is large enough that $m_Y\, Y_Y^{\mathrm{frozen}}$ exceeds $\zeta_{\max}(\tau_Y)$ for *any* lifetime within the BBN-sensitive range. As a consequence, the BBN bound reduces to the simple requirement that $\tau_Y \lesssim 0.1$–$0.2\,\mathrm{s}$, above which charged pions from the mediator's hadronic shower survive long enough to drive $p \leftrightarrow n$ interconversion before thermalizing in the electromagnetic plasma [7]. The BBN constraints associated with longer lifetimes, namely hadrodissociation of ^{4}He ($\tau \sim 10^2$–$10^7\,\mathrm{s}$) and photodissociation ($\tau \gtrsim 10^4\,\mathrm{s}$), are never probed in our parameter space, as the large mediator abundance ensures that the constraint is already saturated at the minimum $p \leftrightarrow n$ interconversion threshold.

Validity and caveats. Two approximations enter the BBN bound as implemented. First, the power-law extrapolation below 30 GeV is conservative, as at lower masses the hadronic activity per decay is suppressed faster than the power law captures, so the true $\zeta_{\max}$ is larger (less constraining) than the extrapolated value and the code does not under-exclude parameter space. The extrapolation above 1000 TeV rests on the asymptotic QCD multiplicity scaling and is better motivated. Second, the bound is applied on each channel independently, weighted by the branching ratio, but leptonic channels carry no hadronic constraint in the short-lifetime regime relevant here, and the remaining hadronic final states (up-type and down-type quarks, gluons, electroweak bosons, hh) all produce similar pion-dominated showers and are grouped under the $u\bar{u}$ curves from [7], so that only the relative weight of $b\bar{b}$ against this $u\bar{u}$-like combination matters. In practice, both approximations have minimal impact on the final result, especially since the large mediator abundance characteristic of the hidden-sector scenarios under consideration cause the constraint to collapse to a single lifetime threshold $\tau_Y \lesssim 0.1$–$0.2\,\mathrm{s}$, which is insensitive to the detailed shape of the $\zeta_{\max}(\tau_Y)$ curve and hence to the interpolation and extrapolation choices above.

5.3 Constraints from Direct Detection

The third quantity of interest is the upper bound on the portal coupling from spin-independent DM–nucleon scattering experiments. At fixed hidden-sector parameters, the SI cross section $\sigma_{\mathrm{SI}}(\varepsilon)$

is a monotonic function of ε through the portal-induced DM–nucleon amplitude discussed in Sec. 3 for each model; its explicit form is Eq. (3.19) for the `VectorPortal`, Eqs. (3.22) and (3.24) for the `BLPortal` and `BaryonPortal`, and Eq. (3.37) for the `HiggsPortal`. The `LiLjPortal` has no tree-level coupling to nucleons and is therefore unconstrained. The code compares $\sigma_{\rm SI}(\varepsilon,\, m_X)$ against the 90% CL upper limit $\sigma_{\rm limit}(m_X)$ from a chosen experiment, with three built-in options: the latest constraints reported by the LZ [8] and XENON [9] collaborations, and a pre-tabulated coherent neutrino scattering floor [10] used for the ultimate reach estimate. Each limit is loaded from the corresponding tabulated data file and evaluated through a log–log interpolation in $(m_X,\, \sigma_{\rm SI})$.

For a given m_X and fixed hidden-sector parameters, the maximum portal coupling compatible with the chosen constraint, $\varepsilon_{\rm DD}(m_X)$, is obtained by a one-dimensional root search on $\log_{10}\varepsilon$ of the condition

$$\sigma_{\rm SI}(\varepsilon_{\rm DD},\, m_X) = \sigma_{\rm limit}(m_X)\,. \tag{5.4}$$

For DM masses outside the validity range of the chosen limit table, the function returns `NaN`; for models with vanishing tree-level cross section, i.e. the `LiLjPortal`, it returns $+\infty$, reflecting the absence of a meaningful direct-detection constraint.

5.4 Thermalization Floor

The hidden-sector freeze-out picture underlying this work assumes that the hidden and visible sectors are not in thermal equilibrium at the time of DM freeze-out. For completeness, the code provides an estimate of the portal coupling above which this assumption breaks down, following the approach of Ref. [2]. The condition is that the mediator decay rate into SM final states exceeds the Hubble rate at the freeze-out temperature $T_{\rm FO} \equiv m_X/x_{\rm FO}$ (with a default $x_{\rm FO} = 20$),

$$\Gamma(Y \to \mathrm{SM}) \geq H(T_{\rm FO})\,, \tag{5.5}$$

which, since all partial widths scale as ε^2, yields the thermalization floor

$$\varepsilon_{\rm therm} = \sqrt{\frac{H(T_{\rm FO})}{\Gamma_{\rm norm}(Y \to \mathrm{SM})}}\,, \tag{5.6}$$

where $\Gamma_{\rm norm}$ is the total mediator decay width to SM particles evaluated at $\varepsilon = 1$. For portal couplings above $\varepsilon_{\rm therm}$ the two sectors equilibrate, so that while DM still annihilates predominantly via $XX \to YY$, both species now share a common temperature with the SM bath and the hidden-sector dynamics that give rise to entropy dilution no longer applies.

A Additional Approximations and Validity

This appendix collects the two main additional assumptions that are common to the entire Boltzmann system and that have not been discussed in the main text, namely the use of Maxwell–Boltzmann (MB) statistics for all hidden-sector species, and the assumption of internal kinetic equilibrium at a single hidden-sector temperature T_h. We then summarize the dominant sources of numerical error and delineate the region of parameter space over which the code has been validated.

Source	Estimated error
ODE integration (Radau, `rtol`=10^{-8})	$\lesssim 0.01\%$ on Y_X
MB statistics vs. full integral ($z \sim 1$–3)	1–10% on $n_{\rm MB}^{\rm eq}$; $\lesssim 1\%$ on $\Omega_X h^2$
g_* interpolation	$\lesssim 1\%$ (smooth tables)
`HDECAY` decay widths	$\lesssim 5\%$ (see Sec. 3.5.2)
BBN bound interpolation	$\lesssim 10\%$ (grid density dependent)
$\Delta_{1,2,3}$ cannibal coefficients	$\mathcal{O}(1)$ uncertainty

Table 3: Dominant sources of numerical uncertainty in the relic abundance calculation. All estimates refer to the validated parameter region described in the text.

Maxwell–Boltzmann statistics. All equilibrium distributions, collision integrals, and the chemical-potential parametrisation $Y_i = Y_i^{\rm eq}\, e^{\bar{\mu}_i}$, Eq. (4.5), rely on MB statistics. The deviations from the full quantum distributions are largest in the relativistic regime, $x_h \equiv m/T_h \lesssim \mathcal{O}(1)$, where both species are still in chemical equilibrium, and become exponentially small by the time freeze-out occurs for $x_h \gtrsim 20 - 30$. As a whole, the MB contributes $\lesssim 1\%$ uncertainty on the computed DM relic abundance $\Omega_X h^2$, which is well below other systematics.

Kinetic equilibrium within the hidden sector. All hidden-sector species are assumed to share a common temperature T_h at all times, so that the phase-space distribution of each species is fully characterised by its mass, chemical potential, and T_h. This requires elastic scattering processes within the hidden sector, such as $XY \to XY$, $XX \to XX$, $YY \to YY$, to remain fast compared to the Hubble rate throughout the evolution. In the models considered here the elastic rates scale as α_X^2 and are parametrically larger than the $2 \to 2$ annihilation and $3 \to 2$ cannibal rates that govern the chemical evolution, so kinetic equilibrium is maintained to a good approximation. The assumption would break down in scenarios with very small hidden-sector couplings or large mass hierarchies, where the elastic scattering rate could drop below H before chemical freeze-out is complete. No explicit check of this condition is performed by the code; it is the user's responsibility to verify that the elastic rates remain sufficiently fast for a particular choice of hidden-sector parameters.

Numerical precision and regions of reduced accuracy. The code has been tested over the parameter ranges $m_X \sim 5\,\text{GeV} - 100\,\text{TeV}$, $m_Y \sim 0.5 - 70\,\text{TeV}$, mass ratios $r = m_X/m_Y$ between 1.3 and 10, and hidden-sector couplings $\alpha_X \gtrsim 10^{-4}$ (or $\lambda_{s,p} \gtrsim 10^{-2}$ for the Higgs portal). Within this region the relic abundance is accurate at the sub-percent level, dominated by the ODE integration error. Outside this range, the most common source of difficulty is either (i) the very late freeze-out which occurs for small hidden-sector couplings ($\alpha_X \lesssim 10^{-4}$) or (ii) the prolonged cannibal equilibrium of the mediator at large mass ratios ($r \gtrsim 10$), which delays its departure from equilibrium over an extended range in x. In both cases the convergence criterion of Eq. (4.18) may not be reached within the default integration range, requiring an increased $x_{\rm max}$ or finer grids. Together with the model-specific uncertainties discussed in the main text, the dominant sources of numerical error and their estimated magnitudes are collected in Table 3.

B Details on the Numerical Implementation and Alternative Solvers

This appendix collects additional technical details on the numerical integration of the Boltzmann system presented in Sec. 4.1. We first describe the alternative solver strategies that complement the primary three-phase solver of Sec. 4.1.2, then summarize the numerical settings common to all of them — including the integration grid, tolerances, and regularization procedures — and finally outline the two-stage scan architecture that combines them to produce the bounds plots presented in this work.

B.1 Alternative solvers

Three additional solvers are implemented alongside the primary three-phase solver of Sec. 4.1.2. Two are intended for production use at portal couplings near or above the thermalization floor, while the third is retained for cross-validation purposes only.

The joint solver (`solve_boltzmann_joint`) extends the three-phase strategy to include the portal-induced $Y \leftrightarrow$ SM exchanges throughout the Boltzmann evolution, as discussed in detail in Sec. 4.2. It modifies the coefficients C and E according to Eqs. (4.24)–(4.25) and splits the intermediate epoch into a dark-locked branch (Phase 1.5-D) and an SM-locked branch (Phase 1.5-S), with the latter entered automatically when the per-particle inverse-decay rate satisfies $\Gamma_Y\, n_Y^{\mathrm{eq,SM}}/(H\, n_Y) \geq 1$. This is the production solver for $\varepsilon \gtrsim 10^{-10}$, where the post-hoc decay correction of Eq. (4.22) would otherwise miss the inverse-decay contribution; below this threshold it agrees with the primary solver at the sub-percent level but is slower because of the additional algebra in the modified coefficients.

The thermal-SM solver (`solve_boltzmann_thermSM`) is the standalone Phase 1.5-S limit of the joint solver. It assumes from the outset that Y is in detailed balance with the SM bath ($\xi = 1$, $\bar{\mu}_Y = 0$) and integrates a single ODE for the comoving DM yield Y_X. This solver is intended for the fully thermalized regime, $\varepsilon \gg \varepsilon_{\mathrm{therm}}$, where it provides a direct check of the joint solver's SM-locked branch and reproduces the standard WIMP freeze-out as a special case.

The two-phase solver (`solve_boltzmann_chempot_2phase`) skips the intermediate Phase 1.5 entirely, transitioning directly from full equilibrium (Phase 1) to the complete three-variable system (Phase 2) when Γ_{ann}/H drops below 10^7. This introduces unnecessary stiffness near the annihilation freeze-out transition, where Y is still effectively in equilibrium but X is departing, precisely the difficulty that the intermediate phase of the primary solver is designed to resolve. It is retained as a cross-check on the three-phase decomposition and is not used in production.

B.2 Numerical settings

All solvers use `scipy.integrate.solve_ivp` with the implicit Runge–Kutta method Radau (L-stable, suitable for stiff problems). The maximum step size is set to 1 in x, with relative tolerance 10^{-8} and absolute tolerance 10^{-15} during Phase 2 (relaxed to 10^{-12} during Phase 1). Rather than integrating from $x_{\min}$ to $x_{\max}$ in a single call, the evolution is broken into segments over a predefined grid of breakpoints,

$$x \in \{10^{-2},\, 10^{-1},\, 0.99\} \cup \mathrm{linspace}(1,\, 10,\, 20) \cup \{10.5,\, 11.5,\, \ldots,\, 99.5\} \cup \{200,\, 500,\, \ldots,\, x_{\max}\}\,, \tag{B.1}$$

with 500 evaluation points per segment. This segmented approach improves the robustness of the solver. The integrator restarts fresh at each breakpoint, which helps it adapt its internal step size to the rapidly changing stiffness of the system. Phase-switching conditions are nevertheless checked continuously within each segment.

Several clipping and regularisation procedures are applied to prevent numerical overflow and underflow. The temperature ratio is restricted to $\ln \xi \in [-20, 20]$ and the dimensionles chemical potentials to $|\bar{\mu}_i| \leq 10/x$ per Hubble time. Exponentials of yields are clipped to the range $[-700, 700]$, and densities that fall below 10^{-300} are replaced with $\pm 10^{-300}$, preserving the original sign.

In addition to the default convergence criterion of the primary solver described in Sec. 4.1.2, an alternative mode is available in which the solver instead monitors the rate of change of the DM yield and terminates when $|d \ln Y_X / d \ln x|$ falls below the configurable tolerance (the parameter `convergence_threshold` in the code). This mode can be selected via the `convergence_mode` option.

C Derivation of the Boltzmann system in Chemical-potential Variables

This appendix provides the step-by-step derivation of the coupled ODE system in Eqs. (4.6)–(4.8) and of the coefficient functions (4.9)–(4.14) summarised in Sec. 4.1. We retain the mediator decay width Γ_Y throughout for completeness, and because the joint evolution that includes it is used for the cross-check discussed in Sec. B. As shown in Sec. 4.2, setting $\Gamma_Y = 0$ is an excellent approximation in the regime of interest ($\varepsilon \lesssim 10^{-10}$, where $\Gamma_Y \ll H$ throughout the freeze-out process), and doing so at any point in the expressions below recovers the system solved by the default Boltzmann integrator.

Starting from the dimensionless chemical-potential parametrisation $Y_i = Y_i^{\rm eq}\, e^{\bar{\mu}_i}$ of Eq. (4.5) and differentiating with respect to $x = m_X/T_{\rm SM}$,

$$\frac{dY_i}{dx} = Y_i \left[\frac{d\bar{\mu}_i}{dx} + \frac{d \ln Y_i^{\rm eq}}{dx} \right] . \tag{C.1}$$

The log-derivative of the equilibrium yield follows from $Y_i^{\rm eq} = n^{\rm eq}_{{\rm MB},\,i}(T_h)/s(T_{\rm SM})$, the relation $\ln T_h = \ln \xi + \ln m_X - \ln x$ (so that $d \ln T_h/dx = d \ln \xi/dx - 1/x$), and SM entropy conservation $d \ln s/dx = -3\tilde{g}/x$, with $\tilde{g}$ defined in Eq. (2.4),

$$\frac{d \ln Y_i^{\rm eq}}{dx} = \lambda_i \left(\frac{d \ln \xi}{dx} - \frac{1}{x} \right) + \frac{3\tilde{g}}{x} , \tag{C.2}$$

where λ_i is given by Eq. (2.7). Combining the two,

$$\frac{dY_i}{dx} = Y_i \left[\frac{d\bar{\mu}_i}{dx} + \lambda_i \left(\frac{d \ln \xi}{dx} - \frac{1}{x} \right) + \frac{3\tilde{g}}{x} \right] , \tag{C.3}$$

which is the identity used repeatedly below to trade dY_i/dx for the variables $d\bar{\mu}_i/dx$ and $d \ln \xi/dx$.

The number-density Boltzmann equations (4.1)–(4.2), supplemented by the mediator decay $Y \to$ SM as a number sink $-\Gamma_Y\, n_Y$ in the Y equation, are converted to comoving yields via

$\dot{Y}_i = (Hx/\tilde{g})\, dY_i/dx$ and $\dot{n}_Y = s\dot{Y}_Y - 3Hs\, Y_Y$, giving

$$\frac{dY_X}{dx} = -\frac{s\,\tilde{g}}{H_{\rm tot}\,x}\,\langle\sigma v\rangle_X \left(Y_X^2 - \frac{(Y_X^{\rm eq})^2}{(Y_Y^{\rm eq})^2}\,Y_Y^2\right), \tag{C.4}$$

$$\frac{dY_Y}{dx} = -\frac{dY_X}{dx} - \frac{\tilde{g}}{H_{\rm tot}\,x}\,\Gamma_{\rm can}\,(1 - e^{-\bar{\mu}_Y}) - \frac{\tilde{g}}{H_{\rm tot}\,x}\,\Gamma_Y\,Y_Y\,. \tag{C.5}$$

The last term in (C.5) is the decay contribution, arising from the $-\Gamma_Y\, n_Y$ sink rewritten as $\dot{Y}_Y\big|_{\rm decay} = -\Gamma_Y\, Y_Y$ and converted to the x-derivative. The hidden-sector energy conservation equation (4.3) also picks up a source term, since each decay removes the average energy B_1^Y from the hidden plasma and deposits it into the SM bath,

$$\dot{\rho}_h + 3H(\rho_h + P_h) = -\Gamma_Y\, B_1^Y\, n_Y\,, \tag{C.6}$$

where we have used $\rho_i = B_1^i\, n_i$ from Eq. (2.8).

Dark-matter and mediator equations. The DM equation is unaffected by Γ_Y. Dividing Eq. (C.4) by Y_X, applying the identity (C.3) on the left-hand side, and expressing the collision integral in chemical potentials via $Y_i = Y_i^{\rm eq}\, e^{\bar{\mu}_i}$, one obtains

$$\frac{d\bar{\mu}_X}{dx} + \lambda_X\left(\frac{d\ln\xi}{dx} - \frac{1}{x}\right) + \frac{3\tilde{g}}{x} = -\Gamma_{\rm ann}\left(e^{\bar{\mu}_X} - e^{2\bar{\mu}_Y - \bar{\mu}_X}\right), \tag{C.7}$$

with $\Gamma_{\rm ann} \equiv (s\,\tilde{g})/(H_{\rm tot}\,x)\,\langle\sigma v\rangle_X\, Y_X^{\rm eq}$. Rearranging in the form $d\bar{\mu}_X/dx = A + B\, d\ln\xi/dx$ directly reproduces Eqs. (4.9)–(4.10) of the main text, with $A = -\Gamma_{\rm ann}(e^{\bar{\mu}_X} - e^{2\bar{\mu}_Y - \bar{\mu}_X}) + (\lambda_X - 3\tilde{g})/x$ and $B = -\lambda_X$.

The mediator equation $d\bar{\mu}_Y/dx = (dY_Y/dx)/Y_Y - d\ln Y_Y^{\rm eq}/dx$ decomposes into three collision contributions plus the redshifting of $Y_Y^{\rm eq}$. The annihilation source, obtained from the $-dY_X/dx$ term in Eq. (C.5), is

$$\frac{1}{Y_Y}\left(-\frac{dY_X}{dx}\right) = \Gamma_{\rm ann}\,\frac{Y_X^{\rm eq}}{Y_Y^{\rm eq}}\,e^{-\bar{\mu}_Y}\left(e^{2\bar{\mu}_X} - e^{2\bar{\mu}_Y}\right). \tag{C.8}$$

The $3 \to 2$ cannibal contribution yields the standard $-(\tilde{g}/x)(\Gamma_{\rm can}/H_{\rm tot})(1 - e^{-\bar{\mu}_Y})$ term already appearing in Eq. (4.11), and the decay term in (C.5) gives simply $-(\tilde{g}/x)(\Gamma_Y/H_{\rm tot})$, independent of $\bar{\mu}_Y$ and of $d\ln\xi/dx$. Subtracting $d\ln Y_Y^{\rm eq}/dx$ via (C.2) and collecting terms gives $d\bar{\mu}_Y/dx = C + D\, d\ln\xi/dx$, with

$$C = \Gamma_{\rm ann}\,\frac{Y_X^{\rm eq}}{Y_Y^{\rm eq}}\,e^{-\bar{\mu}_Y}\left(e^{2\bar{\mu}_X} - e^{2\bar{\mu}_Y}\right) - \frac{\tilde{g}}{H_{\rm tot}\,x}\,\Gamma_{\rm can}(1 - e^{-\bar{\mu}_Y}) - \frac{\tilde{g}}{H_{\rm tot}\,x}\,\Gamma_Y + \frac{\lambda_Y - 3\tilde{g}}{x}\,, \tag{C.9}$$

$$D = -\lambda_Y\,. \tag{C.10}$$

Relative to the $\Gamma_Y = 0$ system of Eq. (4.11), the coefficient C acquires a constant decay drain $-(\tilde{g}/x)(\Gamma_Y/H_{\rm tot})$ that depletes the mediator yield below its chemical-equilibrium value.

Temperature equation. The temperature equation is obtained from the hidden-sector energy conservation law (C.6). Writing $\rho_h = \sum_i s\, B_1^i(T_h)\, Y_i$, the time derivative is

$$\dot{\rho}_h = \dot{s} \sum_i B_1^i\, Y_i + s \sum_i \left[\frac{dB_1^i}{dT_h}\, \dot{T}_h\, Y_i + B_1^i\, \dot{Y}_i \right]. \tag{C.11}$$

Substituting this into Eq. (C.6), using $\dot{s} = -3Hs$, $3H(\rho_h + P_h) = 3Hs \sum_i B_2^i\, Y_i$, and the identity $B_2^i = B_1^i + T_h$ from Eq. (2.9), the B_1^i terms on the two sides cancel. Converting all cosmic-time derivatives to x-derivatives via $\dot{f} = (Hx/\tilde{g})\, df/dx$ and dividing through by $sHx/\tilde{g}$, one finds

$$\frac{dT_h}{dx} \sum_i \frac{dB_1^i}{dT_h}\, Y_i + \sum_i B_1^i\, \frac{dY_i}{dx} + \frac{3\tilde{g}\, T_h}{x} \sum_i Y_i = -\frac{\tilde{g}}{H_{\rm tot}\, x}\, \Gamma_Y\, B_1^Y\, n_Y\,, \tag{C.12}$$

where we have used $H \approx H_{\rm tot}$ in the Hubble-dilution piece on the left, an excellent approximation since the hidden sector is at most a subdominant component of the total energy density at the epoch of interest, while keeping the full $H_{\rm tot}$ in the decay term on the right.

Replacing dY_i/dx with the identity (C.3) and using $dT_h/dx = T_h(d\ln\xi/dx - 1/x)$, the terms proportional to $(d\ln\xi/dx - 1/x)$ combine into the energy denominator $\tilde{D}$ already defined in Eq. (4.15), the terms involving $d\bar{\mu}_i/dx$ give $\sum_i B_1^i\, n_i\, d\bar{\mu}_i/dx$, and the remaining $1/x$ terms combine into $(3\tilde{g}/x) \sum_i B_2^i\, n_i$ after using $B_2^i = B_1^i + T_h$. Equation (C.12) then becomes

$$\tilde{D}\left(\frac{d\ln\xi}{dx} - \frac{1}{x} \right) + \sum_i B_1^i\, n_i\, \frac{d\bar{\mu}_i}{dx} + \frac{3\tilde{g}}{x} \sum_i B_2^i\, n_i = -\frac{\tilde{g}}{H_{\rm tot}\, x}\, \Gamma_Y\, B_1^Y\, n_Y\,. \tag{C.13}$$

Solving for $d\ln\xi/dx$ yields the form $d\ln\xi/dx = E + F\, d\bar{\mu}_X/dx + G\, d\bar{\mu}_Y/dx$ with

$$E = \frac{1}{x}\left(1 - 3\tilde{g}\, \frac{n_X\, B_2^X + n_Y\, B_2^Y}{\tilde{D}} \right) - \frac{\tilde{g}}{H_{\rm tot}\, x}\, \Gamma_Y\, \frac{B_1^Y\, n_Y}{\tilde{D}}\,, \tag{C.14}$$

$$F = -\frac{B_1^X\, n_X}{\tilde{D}}\,, \qquad G = -\frac{B_1^Y\, n_Y}{\tilde{D}}\,, \tag{C.15}$$

which reproduces the coefficients of Eqs. (4.13)–(4.14) with the additional decay contribution to E. Physically, this extra term accelerates the cooling of the hidden sector as decaying mediators carry energy out of the dark plasma while depositing entropy into the SM bath, driving T_h down faster than Hubble expansion alone.

Decoupled system. Substituting $d\bar{\mu}_X/dx = A + B\, d\ln\xi/dx$ and $d\bar{\mu}_Y/dx = C + D\, d\ln\xi/dx$ into the expression for $d\ln\xi/dx$ yields the decoupled master equation (4.8), with the chemical-potential equations then following by back-substitution through (4.6)–(4.7). The algebraic structure of this decoupling is identical to the $\Gamma_Y = 0$ case; all decay dependence is absorbed into the modified coefficients C and E given in Eqs. (C.9) and (C.14). Specifically, C acquires a constant drain $-(\tilde{g}/x)(\Gamma_Y/H_{\rm tot})$ that depletes the mediator yield, and E acquires an energy-loss term $-(\tilde{g}/x)(\Gamma_Y/H_{\rm tot})(B_1^Y n_Y/\tilde{D})$ that accelerates the cooling. The coefficients A, B, D, F, G and the energy denominator $\tilde{D}$ are unchanged, and in the limit $\Gamma_Y \ll H_{\rm tot}$ both corrections vanish, recovering the system solved by the default Boltzmann integrator.

D Physical Constants

Table 4 collects the numerical values of all physical constants used in the code, taken from the Particle Data Group [11].

Symbol	Value	Description
Fundamental constants		
$M_{\rm Pl}$	$2.435 \times 10^{18}\,\text{GeV}$	Reduced Planck mass
v_H	$246.22\,\text{GeV}$	Higgs vacuum expectation value
m_N	$0.9389\,\text{GeV}$	Nucleon mass
$\hbar^2 c^2$	$0.3894 \times 10^{-27}\,\text{GeV}^2\,\text{cm}^2$	Conversion factor
$\hbar$	$6.582 \times 10^{-25}\,\text{GeV} \cdot \text{s}$	Reduced Planck constant
Electroweak parameters		
$\sin^2\theta_W$	0.23121	Weinberg angle
$\alpha_{\rm EM}$	1/137.0	Fine-structure constant
e	0.3029	EM coupling
g_W	0.6520	Weak coupling
$g_Y^{\rm SM}$	0.3577	Hypercharge coupling
α_s	0.1179	Strong coupling at M_Z
Boson masses		
m_h	$125.10\,\text{GeV}$	Higgs mass
M_Z	$91.1876\,\text{GeV}$	Z boson mass
M_W	$80.379\,\text{GeV}$	W boson mass
Γ_Z	$2.4952\,\text{GeV}$	Z total width
Fermion masses		
m_e	$0.511 \times 10^{-3}\,\text{GeV}$	Electron
m_μ	$0.1057\,\text{GeV}$	Muon
m_τ	$1.777\,\text{GeV}$	Tau
m_u	$2.16 \times 10^{-3}\,\text{GeV}$	Up quark
m_d	$4.67 \times 10^{-3}\,\text{GeV}$	Down quark
m_s	$93.4 \times 10^{-3}\,\text{GeV}$	Strange quark
m_c	$1.27\,\text{GeV}$	Charm quark
m_b	$4.18\,\text{GeV}$	Bottom quark
m_t	$172.69\,\text{GeV}$	Top quark
Cosmological observables		
s_0	$2891\,\text{cm}^{-3}$	Entropy density today
ρ_c	$1.054 \times 10^{-5}\,\text{GeV}\,h^2\,\text{cm}^{-3}$	Critical density
$\Omega_c h^2$	0.12	Observed DM relic density
$m_Y^{\rm relic}$	$\Omega_c h^2\,\rho_c/s_0$	Reference relic yield $\times$ mass

Table 4: Physical constants and Standard Model parameters used throughout the code.

E Code Architecture

The code is organized around a strict separation between physics and numerics. Model instances (`VectorPortal`, `HiggsPortal`, etc.) are stateless, carrying only their physical parameters and providing rate methods with no reference to a `Cosmology` instance. The `BoltzmannSolver` combines a `Cosmology` and a `DarkSectorModel` instance, reading all model-specific physics through the abstract interface. Mass scans are handled via a factory pattern in which `find_mX_DMRelic_joint` takes a callable that creates model instances for different mass points. Heavy initializations such as `HDECAY` compilation and BBN data loading are deferred until first use and cached thereafter. The full module layout is shown below.

```
hidden_sector_DM/
  __init__.py              Public API exports
  constants.py             Physical constants, fermion tables
  cosmology.py             Cosmology class (g*, Hubble, thermo)
  model.py                 HiddenSectorModel ABC + 5 portals
  solver.py                BoltzmannSolver (3phase, joint, thermSM,
                           2phase, background) +
                           find_epsilon_*, find_mX_DMRelic_joint
  bbn.py                   BBN hadronic injection bounds
  direct_detection.py      LZ, XENONnT, neutrino floor interpolators
  HiddenSectorDM.py        Backward-compatible re-exports

  phi_decay/
    __init__.py            ensure_hdecay_ready, public API
    phi_decay_hybrid.py    Three-regime stitching
    hdecay_interface.py    HDECAY Fortran wrapper
    scalar_portal/         Low-mass regime (< 5 GeV)
      api/model.py         LightScalar / HeavyScalar
      decay/               Decay channel implementations
      data/                Form factors, QCD, constants

  data/
    direct_detection_data/ LZ, XENONnT, neutrino floor tables
    bbn_constraints_decay/ 12 BBN data files

python/
  get_*PDM_bounds_all.py   Two-stage bounds scans (per portal)
  run_relic.py             Single-point relic-abundance runner
  branching_ratios_pie.py  Mediator branching-ratio plots
```

AI Disclosure

The initial draft of this document was generated by Claude (Anthropic) from an analysis of the codebase. The final version has been extensively rewritten and edited by the author; very little of the original machine-generated text remains.

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